

$$\begin{array}{|l}
\text{cumulative_total} : \text{seq } \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
\text{cumulative_total}(\langle \rangle) = 0 \wedge \\
(\forall x : \mathbb{N}; s : \text{seq } \mathbb{N} \bullet \\
\text{cumulative_total}(\langle x \rangle \frown s) = x + \text{cumulative_total}(s))
\end{array}$$